5 Airway Management & Oxygen Therapy

HME922 Procedural Station · 10 minutes · Airway Management & Oxygen Therapy guideline

guideline wording — say it aloud DOPS On the examiner's DOPS marking form ☐ tick when done

KEY MESSAGE

Work through the **4 O's**: **Observe** & clear → **Open** → **Optimise** with adjuncts → **Oxygenate** to a **documented SpO2 target**.

UNIVERSAL OPENING

- ☐ Hand hygiene — Moment 1
- ☐ Apply **PPE**
- ☐ Introduce yourself: name + "medical student"
- ☐ Confirm ID — **3 identifiers**
- ☐ Explain purpose of the assessment / oxygen
- ☐ Know the history (e.g. **COPD** → 88–92%); ask allergies (latex, tape)
- ☐ Allow questions; **informed consent** (unconscious = implied)
- ☐ Sit patient up if possible; warm, comfortable; privacy, lighting
- ☐ Stethoscope + airway equipment ready

MEMORY AID — THE 4 O's

Observe (FB, teeth, blood, vomit) → **O**pen (chin lift, head tilt, jaw thrust) → **O**ptimise (suction, OPA, NPA, LMA) → **O**xygenate (prongs, Hudson, NRB, BVM)

5 Moments of Hand Hygiene: 1 before touching patient · 2 before procedure · 3 after body-fluid risk · 4 after touching patient · 5 after touching surroundings

CLINICAL KNOWLEDGE — SAY IT

- ☐ Goal of O2: correct **alveolar and/or tissue hypoxia**
- ☐ Indications: respiratory distress · hypoxaemia · unwell/deteriorating incl. trauma · **SaO2 < 93%** · surgery · opiates/sedatives
- ☐ Targets: **92–96%** acutely ill · **88–92%** COPD / chronic respiratory failure
- ☐ Nurses may start O2 without an order, but follow the **target range documented by medical staff**; oximetry for all on O2; ABG/VBG if breathless + acutely ill
- ☐ Rising O2 need or no response = **deterioration** → urgent review, escalate

EQUIPMENT

- ☐ Stethoscope · torch · **Magill forceps** · suction + **Yankauer**/catheters · OPAs · NPAs · LMA/SGA · O2 wall/cylinder · **white flowmeter** · tubing · delivery devices

O1 — OBSERVE & CLEAR

- ☐ Look for foreign body, loose teeth, blood, vomit, saliva
- ☐ Mild obstruction: sit up, cough, monitor, send for help
- ☐ Severe: forceps with caution · lean forward + cough · back blows / chest thrusts · suction

O2 — OPEN

- ☐ Head tilt–chin lift
- ☐ **Suspected C-spine injury → jaw thrust ONLY**

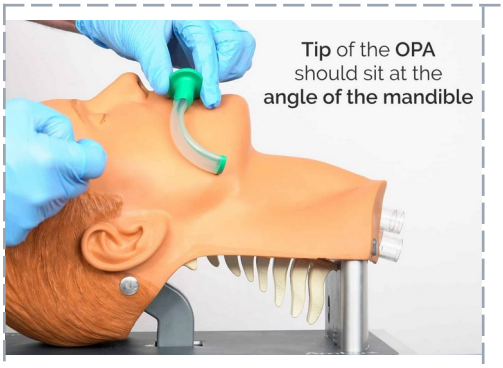
O3 — OPTIMISE WITH ADJUNCTS

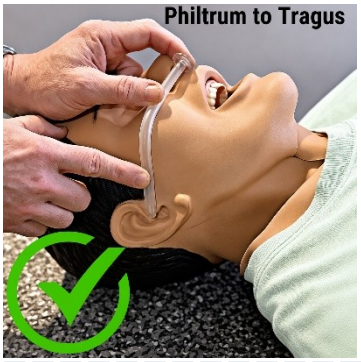
Adjunct	Who	Size	Key points
OPA (Guedel)	**Unconscious only**	Centre of top teeth → angle of mandible	Follow natural curve. Risks: aspiration, mouth/teeth trauma, wrong size → obstruction / laryngospasm
NPA	Awake or unconscious; better tolerated awake + children	Nares → tragus	Safety pin, lubricate, along nasal floor perpendicular to face, flange at nostril, tie with trache tape. **NO: basal skull # · mid-face # · coagulopathy**; caution septum/polyps
LMA / SGA	Unconscious; alternative to intubation	By weight; women 3–4, men 4–5	Lubricate, slide along back of throat until it stops, inflate cuff if needed, tape. **NO: aspiration risk**
Suction	Any	—	Wall outlet, canister, tubing, Yankauer

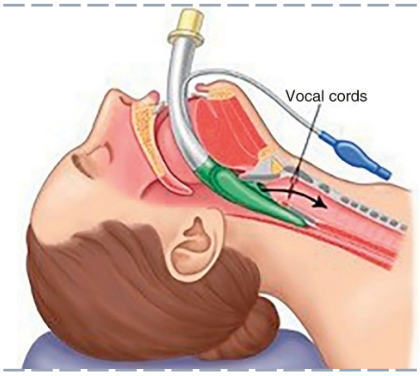
MEMORY AID — SIZING

OPA = Teeth → Jaw ("mouth tube, measure the mouth") · **NPA = Nose → Ear** ("nose tube, measure the nose")

OPA in an awake patient = gag/vomit → use an NPA.







BREATHING ASSESSMENT

- ☐ **LOOK** chest rise · **LISTEN** air at mouth/nose · **FEEL** air movement
- ☐ Observe **SpO2 · respiratory rate · colour**; perfusion, work of breathing

O4 — OXYGENATE

Device	Flow	FiO2	Key points
Nasal prongs	**1–4 L/min**	24 / 28 / 32 / 36%	≥ 5 L dries mucosa, crusting, epistaxis. Mouth breathing OK unless nares blocked
Hudson mask	**≥ 5 L/min**	6 L 45% → 10 L 65% (+5% per L)	Min 5 L stops CO2 rebreathing; FiO2 depends on flow, fit, effort; can humidify
Non-rebreather	**≥ 15 L/min**	60–90%	Emergency; spontaneous breathing. **Fill bag > 2/3 first.** No flow = **suffocation risk**
BVM	**15 L/min**	Up to 100%	Not breathing / inadequate breathing; self-inflating

MEMORY AID — FLOW RATES — "4 – 5 – 15 – 15"

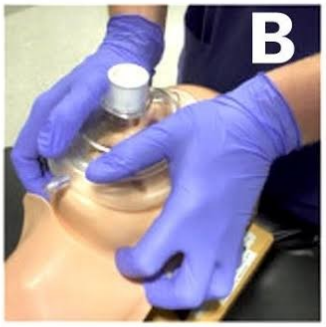
Prongs up to **4** · Hudson at least **5** · NRB **15** · BVM **15**

Prongs FiO2 = **20 + 4 × L** (1 L 24% … 4 L 36%) · Hudson: **6 L = 45%, add 5% per litre** to 10 L = 65%





Two-handed C-E grip



Two-handed vice grip

BVM TECHNIQUE

1. Mask: pointed end on bridge of nose, round end between lip and chin; tight seal
2. **C-E grip:** thumb + index = "C" on mask; 3 fingers = "E" lifting jaw into mask
3. Squeeze **1/3 of the bag**, **10–12 breaths/min**, watch chest rise
4. Over-inflation → gastric insufflation, volutrauma, barotrauma

PULSE OXIMETRY

- ☐ Normal **97–99%**; 93–97% may be normal for some; read as a **trend** with colour, perfusion, work of breathing
- ☐ False readings: cold peripheries / hypovolaemia · severe hypotension · cardiac failure · arrhythmia · poor probe · **CO poisoning** · nail polish

UNIVERSAL CLOSING

- ☐ Used adjuncts, catheters, masks into correct waste
- ☐ Area clean; patient sitting up, comfortable; bed lowest
- ☐ Simple explanation of findings; reassure
- ☐ Call bell — "**buzz** if you're concerned"
- ☐ Hand hygiene — Moments 4/5
- ☐ Document device, flow, SpO2, RR, target range, response
- ☐ Can't answer? "I'll forward that to someone senior"
- ☐ Examiner asks **"How do you think that went?"** → one thing done well · one to improve · one strategy

MY NOTES